EP 12 the Wheatstone Bridge and Specific Resistance

1. Objective

1. To understand the principle of measuring resistance with a Wheatstone Bridge.
2. To measure unknown resistances using a self-built Wheatstone Bridge and a portable Wheatstone Bridge respectively and compare the results.

2. Principle

The core is to compare an unknown resistance with known ones, based on the **bridge balance condition**.

2.1 Core Structure & Balance Condition

The bridge consists of four resistors (R_1, R_2, R_s - known standard resistance, R_x - unknown), a galvanometer (G), and a DC supply.

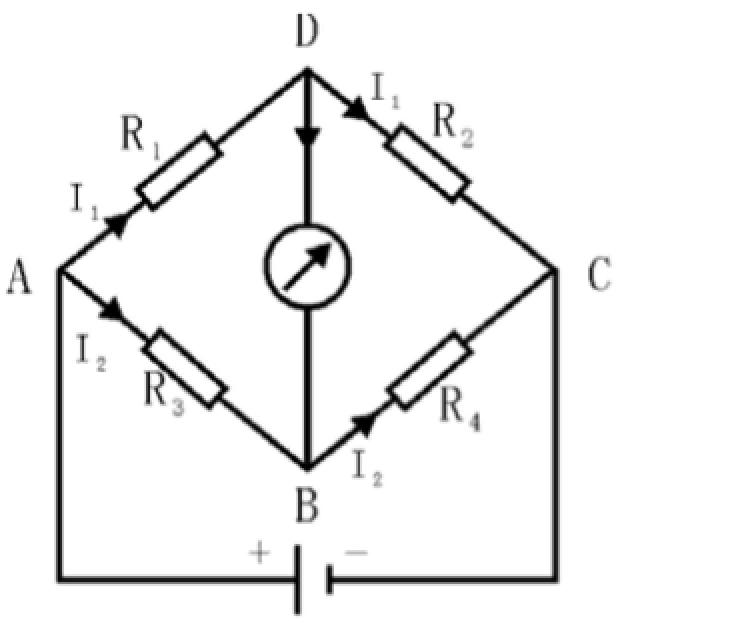

Fig. 1 the Wheatstone Bridge

At balance:

- No current flows through G (zero deflection), so the potential drop across R_1 equals that across R_s, and across R_2 equals that across R_x.

- Key equations:
  $\frac{R_1}{R_2} = \frac{R_s}{R_x}$
  $R_x = \frac{R_1}{R_2} R_s$
  (Ratio $K = \frac{R_1}{R_2}$ is usually an integral power of 10 for easy calculation.)

2.2 Exchange Method for Error Reduction

Errors from R_1 and R_2 (standard resistance boxes) can be minimized by:

1. Keeping K unchanged, swapping R_x and R_s, adjusting R_s to a new balanced value R_s'.
2. Final formula (eliminates R_1/R_2 dependence):
   $R_x = \sqrt{R_s R_s'}$

2.3 Portable Wheatstone Bridge (QJ23)

- Integrates galvanometer, dry cells, and pre-calibrated resistors for convenience.
- Set K (0.001–1000, integral power of 10) via a ratio dial; set R_s via four dials (1000Ω, 100Ω, 10Ω, 1Ω).
- Calculate $R_x = K\ \text{total} R_s$ (when G shows zero deflection).

3. Procedure

1. **Circuit Connection**: Assemble the bridge as shown in the figure: include R (slide-wire resistor for current regulation), R (protective resistor), $R_1/R_2/R_s$ (standard resistance boxes), and R_x (unknown resistor). Confirm the circuit with the instructor before turning on the power.

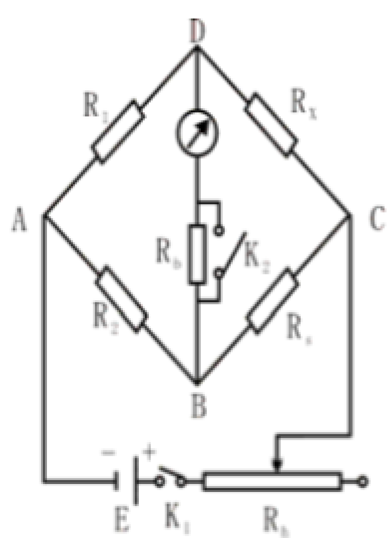

Fig. 3 self-built bridge measure resistances

2. **Balance Adjustment**:

- Set ratio $K = R_1/R_2$ (e.g., $K = 1$ for 100Ω unknown, $K = 1$ for 200Ω unknown) and initial R_s (estimated based on R_x's nominal value to ensure 4 significant figures).
- Close switch K_1 (battery switch); keep K_2 (protective resistor switch) open, then gently press K (galvanometer switch). Adjust R_s step-by-step: if G deflects right, increase R_s; if left, decrease R_s until G shows no deflection.
- Close K_2 to short-circuit R, fine-tune R_s for precise balance, then record R_s.

3. **Exchange Method**: Swap the positions of R_x and R_s, repeat Step 2 to obtain the new balanced R_s', and record the value.
4. **Repeat for Another Resistor**: Replace R_x with the second unknown resistor, repeat Steps 2–3 to collect data.
5. **Comparison with portable Wheatstone bridge measuring resistances**: Measure the same resistances by QJ23 potable Wheatstone bridge and compare the portable bridge's R_x results with those from the self-built bridge.

4. Data Recording and Processing

4.1 Data Form

Table 1:

Nominal values						
R_1/R_2						
Resistances	R_s	R_s'	R_x	R_s	R_s'	R_x

Resistances of potable Wheatstone bridge:

$R_{1x} =$, $R_{2x} =$

4.2 Data Calculation

1. **Self-built Bridge Results**:
   - Unknown 1: R_x
   - Unknown 2: R_{2x}
2. **Portable Bridge Results**:
   - Unknown 1: R_x
   - Unknown 2: R_{2x}
3. **Uncertainty Estimation**:

- Main error source: Instrument precision (±0.01Ω for resistance boxes/dials).
- Relative uncertainty: ≈0.02%; Absolute uncertainty: ≈0.02Ω (100Ω)、≈0.04Ω (200Ω).

4. **Final Results**:
   - Unknown 1: $R_x = R\ _R$
   - Unknown 2: $R_{2x} = R\ _R$

4.3 Data Analysis

1. **Validity Check**:
2. **Consistency**:
3. **Exchange Method Effect**:

5. Error Analysis

5.1 Systematic Errors

- **Resistance Box Precision**: Standard boxes have inherent errors (±0.01Ω), mitigated by the exchange method but not fully eliminated.
- **Contact Resistance**: Loose terminals in the self-built bridge introduce ~0.005Ω extra resistance, causing minor balance deviations.
- **Ratio Dial Offset**: Portable bridge's ratio dial may have tiny offsets (~0.001), leading to ~0.1Ω bias for 200Ω.

5.2 Random Errors

- **Galvanometer Sensitivity**: Low-sensitivity G (≈5μA/division) misses small currents, causing ±0.01Ω R_s deviations.
- **Human Adjustment**: Manual dial adjustments have ~0.5s reaction delays, leading to ±0.005Ω fluctuations.
- **Power Fluctuations**: DC supply voltage varies ±0.1V, affecting current stability and G deflection.

6. Prequestions

1. **Why press B before G in the portable bridge?**
   Pressing B first stabilizes circuit current; pressing G afterward avoids sudden current surges (from circuit transients) that could damage G.

2. **How does $K = R_1/R_2$ affect R_x measurement?**
   Choosing K as an integral power of 10 simplifies calculations. For large R_x, a small K (e.g., 0.01) keeps R_s within dial ranges, ensuring 4 significant figures.